PHOENIX DIRECT AIR CAPTURE

Company Plan & Technical Reference

AI Optimizer · Carbon Ledger · MRV Registry · Cash Flow Model

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Solar PV + LFP battery · Hybrid sorbent-solvent contactor
Mineral carbonation into Arizona copper mine tailings aggregate
Net-negative accounting: gross CO■ minus ALL lifecycle emissions

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1. Executive Summary

Phoenix Direct Air Capture is developing a **permanently net-negative carbon removal business** in the Sonoran Desert. The system captures atmospheric CO■ at night using solar-charged batteries to power contactor fans, releases it by day through solar thermal regeneration at 80–100°C, and sequesters it permanently via mineral carbonation into construction aggregate made from Arizona copper mine tailings — or via direct geologic injection. No fossil energy enters the loop at any point.

A fully working AI optimization and carbon accounting platform has been designed, built, and validated. It comprises 11 Python modules, an 11-tab Streamlit operations dashboard, and a complete suite of lifecycle tools — from a 24-hour linear program dispatcher to a batch-traceability MRV registry, Monte Carlo LCA uncertainty analysis, and a 96-month cash flow model. All tools are running locally today.

114.3 t	88.4%	$285	Month 25	$8,187
Gross CO■	Net-negative ratio	Cost / net tonne	Breakeven	Peak cash need
per year (50 kW pilot)	*P50 range 85.8–92.1%*	*base case, diesel grinder*	*with grants / Month 42 commercial*	*maximum cumulative deficit*

The three core claims — net-negative removal, permanent sequestration, and honest cost-per-net-tonne accounting — are verified continuously by the ledger and MRV registry, not asserted from a spreadsheet. Every kilogram of CO■ credited traces to a specific capture event, a TCLP-qualified tailings lot, a confirmed permanent fate, and a closed mass balance.

Note: This document is pre-commercial and confidential. All performance figures are from prototype simulation with calibrated synthetic Phoenix weather data. Phase 1 bench-scale testing has not yet begun.

2. The Company Plan — Mission & Strategy

2.1 Mission

Capture the air. Turn it into rock. Build with it. The mission is to remove atmospheric CO■ permanently and verifiably, using only the sun as an energy source and Arizona copper mine tailings as the sequestration medium — converting an environmental liability into a structural material.

2.2 Why Phoenix, Why Tailings

- **Solar resource:** Phoenix averages 299 sunny days per year and leads the continental US in GHI. A 50 kW array provides sufficient energy for night capture (via battery) and daytime thermal regeneration — the entire energy loop closes on solar.
- **Tailings supply:** Arizona is the largest copper-producing state in the US. Morenci, Bagdad, Sierrita, Ray, and Pinto Valley generate a combined tailings output exceeding 200 million tonnes per year. Current tailings management is an ongoing environmental liability — reactive silicate tailings that carbonate also benefit from structural stabilisation, creating a mutual value proposition for offtake agreements.
- **Construction market:** Arizona's population growth (Phoenix metro: #1 fastest-growing in the US 2023–2025) creates sustained demand for aggregate and precast concrete. Carbonated aggregate commands a premium from sustainability-focused developers.
- **No fossil inputs:** Phoenix's solar irradiance and the proximity of mine tailings eliminate any need for fossil fuel in the process loop, which is required to claim a genuinely permanent net-negative credit.

2.3 Competitive Differentiation

Dimension	Phoenix DAC	Typical DAC operator
Energy source	Solar PV + LFP battery only	Grid (fossil mix) or natural gas
Sequestration fate	Mineral carbonation (permanent) or geologic	Often EOR or utilisation (not permanent)
Accounting metric	Cost per NET tonne (honest)	Cost per GROSS tonne (inflated 15–30%)
Tailings use	Per-batch geochemistry assay, TCLP-qualified	Fixed stoichiometry assumptions
MRV	Automated batch traceability chain	Manual spreadsheet reporting
Location	Mine tailings adjacent, solar belt	Remote sites, grid-dependent

2.4 Capital-Light Model

The founder is proceeding capital-light: grant and partnership funded, no self-funded plant, operating alongside full-time work and school. Phase 0 requires no company formation — only literature access, patent attorney consultation, and this software. Each phase gates spending on the phase before it: no Phase 1 bench hardware unless Phase 0 freedom-to-operate check passes.

3. Phase Gate Roadmap (Phase 0 → 4)

Each phase only advances when all gates pass. Gates are evaluated live by the PhaseGateEngine in the dashboard — some are automated (pulled from the carbon ledger and MRV registry), others require human confirmation (patent search, ASTM results, etc.).

Phase 0 — Foundation & Literature (6 months)

Phase 1 — Bench Scale & Sorbent Qualification (12 months)

Phase 2 — Pilot Plant (1 t CO■/day) (18 months)

Phase 3 — Commercial Scale (10 t CO■/day) (24 months)

Phase 4 — Scale-Out & Replication (36 months)

Phase	Duration	Key Gates	Funding Source
Phase 0 Foundation & Literature	6 mo	• Systematic literature review (≥50 papers, 2015–present) • Freedom-to-operate (FTO) patent search — hybrid sorbent-solvent contactors • LCA draft v0 (this codebase IS the draft LCA model) • Identify university / lab partner for sorbent testing	Literature access ($0), IP attorney consultation (~$1,500), cloud compute (~$500)
Phase 1 Bench Scale & Sorbent Qualification	12 mo	• Build bench-scale contactor (single column, ~0.1 t CO■/month) • Sorbent capacity ≥ 85% after 500 cycles (GATE: automated, measured) • First TCLP-qualified tailings source from accredited lab • XRD or TGA confirmation of carbonate mineral formation	DOE SBIR Phase I (~$150k / 30 months). Bench equipment: ~$25k. Sorbent: ~$6k/yr.
Phase 2 Pilot Plant (1 t CO■/day)	18 mo	• Net-negative ratio ≥ 85% (GATE: live from ledger) • Cost/net tonne ≤ $600 (GATE: live from ledger) • ASTM C33 aggregate specification passed (GATE: lab test) • 45Q tax credit eligibility confirmed with tax counsel	DOE SBIR Phase II (~$1.5M / 24 months). Pilot plant: ~$100k CapEx. 0.5 FTE ops.
Phase 3 Commercial Scale (10 t CO■/day)	24 mo	• Cost/net tonne ≤ $400 (GATE: live from ledger) • ≥ 2 TCLP-qualified tailings sources (supply chain resilience) • Third-party MRV verification — full year, signed by DNV or SCS Global • Multi-year aggregate offtake agreement signed (revenue ≥ 50% of OpEx)	DOE Prize / AZ State grants (~$100k/yr). Commercial plant: ~$250k CapEx. 1.5 FTE.
Phase 4 Scale-Out & Replication	36 mo	• Cost/net tonne ≤ $300 (GATE: live from ledger) • Net-negative ratio ≥ 88% (GATE: live, with solar crusher) • ≥ 1 solar crusher contract active (GATE: automated) • Replication-ready plant design with BOM validated against actuals	Self-funded from operations. Optional Series A if replication accelerates.

4. Technical Architecture — 11 Python Modules

All code is Python 3.12 with four runtime dependencies: scipy, pandas, plotly, streamlit. Every module is independently containerisable. The 11-tab Streamlit dashboard provides a live view of the full system state without requiring direct code access.

Module	Purpose	Key functionality
`weather.py`	**Synthetic Phoenix weather generator**	8,760-hour synthetic year calibrated to NREL TMY3. Seasonal temperature profile, monsoon humidity (Jul–Sep), haboob dust events, PV output with temperature derating. Replace with live NOAA + watsonx.ai Granite-TTM forecasts in Phase 2.
`sorbent.py`	**Amine sorbent degradation model**	Cycle-by-cycle capacity tracking (0.045%/cycle base rate, accelerated by humidity >40% and regen T >95°C). Temperature-dependent capture capacity (72% at 35°C vs fresh at 25°C). Dust fouling accumulation and reversible cleaning model. Economic replacement trigger.
`tailings.py`	**Per-batch tailings assay engine**	TailingsAssay dataclass with Bond's Law grinding energy, stoichiometry from reactive fraction, transport emissions. 6 pre-loaded Arizona mine defaults. blend_assays() for on-site stockpile mixing. rank_batches() composite scoring. Solar crusher contract cost model.
`ledger.py`	**Net-negative carbon ledger**	Per-hour signed ledger debiting: fan electricity (solar lifecycle 20 g CO■e/kWh), vacuum, battery manufacturing, sorbent manufacturing, filter, tailings grinding+transport, infrastructure. TailingsAssay-aware — uses actual batch geochemistry. Reports cost per GROSS tonne and per NET tonne separately.
`optimizer.py`	**24-hour LP dispatch scheduler**	scipy HiGH linear program maximising net CO■/$ over a 24-hour window subject to battery SOC constraints. 5 additional optimisations: assay-aware capture threshold, dynamic regen temperature (82–95°C by sorbent age), proactive cleaning scheduling, battery-forecast LP re-solve at day-start, greedy fallback if scipy absent.
`mrv.py`	**MRV — Measurement, Reporting & Verification**	Full batch traceability: CaptureBatch → TailingsLotQualification (TCLP) → CarbonationBatch (mass balance ±5%) → ProductBatch (fate gate) → VerifiedCredit. EPA TCLP limit auto-check on all 8 regulated analytes. ASTM C33 gate for aggregate sales. Water ledger per batch. JSON export for auditor handoff.
`lca.py`	**LCA boundary model (ISO 14064-2)**	Full Scope 1/2/3 boundary. Every emission factor cited with source and confidence level. Low-confidence items flagged for Phase 1 improvement. Diesel vs solar crusher comparison. Water accounting with Phoenix stress-index disclosure. confidence_flags() method returns prioritised improvement list.
`phase_gate.py`	**Phase gate engine**	Go/No-Go evaluator for all 5 phases. Automated gates pull live data from ledger, MRV, and sorbent state. Manual gates (FTO check, ASTM, 45Q, etc.) set via dashboard checkboxes. next_actions() returns prioritised remediation list. Prevents phase advancement until all gates pass.
`cashflow.py`	**96-month cash flow model**	Conservative monthly projections for Phase 0→4. Revenue: aggregate sales, voluntary carbon credits, §45Q utilisation ($130/t), grants. Costs: CapEx (annualised), OpEx, sorbent, filters, water, tailings processing, MRV audit, legal. Breakeven analysis with-grants vs commercial-only. Peak cash need calculation.

Module	Purpose	Key functionality
`sensors.py`	**Sensor data pipeline**	16 physical sensor channels with range validation and drift detection. State machine auto-detects capture cycles and reactor runs, committing batches to the MRV registry without manual data entry. co2_captured_kg() from differential concentration × air flow. CSV import + downloadable template. IBM App Connect integration path.
`monte_carlo.py`	**Monte Carlo LCA sensitivity**	500 simulations, pure Python stdlib. 6 uncertain parameters sampled from triangular distributions (literature-sourced bounds). P5/P50/P95 net-negative ratios, probability of missing the 85% gate, tornado chart (Pearson correlations). Key result: diesel P50 = 89.7%, P(below 85%) = 3.6%, top driver: sorbent manufacturing EF.

4.1 Dashboard — 11 Tabs

Tab	Contents
■ Dashboard	KPIs, net-negative gauge, emission donut chart, monthly bar chart
■ Tailings Assay	Editable batch library, derived geochemistry, batch ranking, blend UI
■ Hourly Dispatch	4-panel timeline: mode/temp, battery SOC/PV, CO■ kg/h, dust factor
■ Sorbent Health	Degradation curve, replacement forecast, economic trigger
■ MRV Registry	Verified credits, TCLP results, water accounting, traceability chain
■■ LCA Boundary	ISO 14064-2 breakdown, diesel/solar comparison, confidence flags
■ Phase Gates	Go/No-Go per phase, manual checkboxes, next actions list
■ Cash Flow	96-month projections, stress-test sliders, breakeven chart
■ Sensor Pipeline	Live sensor ingestion, anomaly detection, auto-MRV commit
■ LCA Monte Carlo	Net-neg distribution histogram, tornado chart, uncertain parameter table
■ Export	Hourly CSV download, MRV registry JSON for auditor handoff

5. Carbon Accounting Methodology

5.1 LCA System Boundary (ISO 14064-2)

Functional unit: **1 tonne of net CO■ removed and permanently sequestered.** The boundary includes all Scope 1, 2, and 3 upstream emissions. Scope 1 = zero (no on-site combustion). Scope 2 = zero (100% solar). Scope 3 upstream covers PV manufacturing, battery manufacturing, sorbent manufacturing, filter manufacturing, tailings grinding and transport, and annualised infrastructure. End-of-life for carbonated aggregate is excluded under ISO 14064-2 cut-off rules — the CO■ is locked in stable carbonate mineral.

Emission category	Scope	kg CO■e/yr	% of total	Confidence
Sorbent Mfg	3-upstream	5333.3	48.0%	LOW ■
Tailings Transport	3-upstream	3069.0	27.6%	High
Tailings Grinding	3-upstream	1069.2	9.6%	High
Fan Electricity	Operational	876.0	7.9%	High
Battery Mfg	3-upstream	280.0	2.5%	Medium
Infrastructure	3-upstream	228.6	2.1%	LOW ■
Filter Mfg	3-upstream	90.0	0.8%	LOW ■
Vacuum Electricity	Operational	87.6	0.8%	High
Pv Manufacturing	3-upstream	60.0	0.5%	High
Reactor Electricity	Operational	17.0	0.2%	High
Water	Operational	0.7	0.0%	Medium
TOTAL		**11111**	**100%**	

Net CO■ removed: 103.2 t/yr | Net-negative ratio: 90.3% | Key finding: tailings grinding is 63–67% of all system emissions on diesel crusher. Solar crusher contract is the single highest-leverage improvement.

5.2 Monte Carlo LCA Uncertainty

500 simulations were run sampling 6 uncertain emission factors from triangular distributions (literature-sourced bounds). Results for the **diesel crusher scenario**:

Metric	Value
P5 net-negative ratio	85.8%
P50 net-negative ratio	89.7%
P95 net-negative ratio	92.1%
P(ratio < 85% gate)	3.6%
P50 cost / net tonne	$281
Top uncertainty driver	Sorbent manufacturing EF (4–15 kg CO■e/kg)

The 4% probability of missing the 85% net-negative gate is **driven entirely by sorbent EF uncertainty**. Obtaining a supplier Environmental Product Declaration in Phase 1 would collapse this risk. The solar crusher scenario reduces

P(below 85%) to near zero by eliminating diesel grinding emissions.

5.3 MRV Traceability Chain

Every verified removal credit traces through five records, all of which must be present and passing before a credit is issued:

Step	Record	Key gate
1	CaptureBatch	Power source = solar/battery only; no grid fossil input
2	TailingsLotQualification	All 8 EPA TCLP analytes below regulatory limits
3	CarbonationBatch	Mass balance (tailings + CO■ ≈ aggregate) within ±5%
4	ProductBatch	Fate = sold_aggregate or geologic_injected (NOT stockpiled)
5	VerifiedCredit	TCLP ✓ + ASTM C33 ✓ (for aggregate) + mass balance ✓

Demo registry: 2 credits issued, 2 lots qualified, 1 lot rejected (arsenic exceedance — Sierrita pilot), net water: 9 L consumed (138 L/t CO■).

6. Validated Performance Results

The following results are from a full 365-day simulation with calibrated synthetic Phoenix weather data. Configuration: 50 kW peak PV, 200 kWh LFP battery, 18 kW fans, 2,000 kg sorbent, Morenci Standard Lot tailings (20% reactive, 230 km, diesel grinder), mineral carbonation pathway, $200/tonne CO■ credit price.

114.3 t	101.1 t	13.2 t	88.4%	$252	$285
Gross CO■	Net CO■	System emissions	Net-negative	$/gross tonne	$/net tonne
captured/year	*removed/year*	*CO■e/year total*	*ratio (gross basis)*	*vendor metric*	*honest metric*

6.1 Optimisation Impact (vs Greedy Baseline)

Optimisation	Mechanism	Projected gain
24h LP scheduler	scipy HiGHS LP, maximise net CO■/$	+8–12% net CO■/yr
Assay-aware threshold	Capture threshold scales with batch quality	+3–5% net-neg ratio
Dynamic regen temp	82–95°C by sorbent age (saves 15% sorbent life)	−15% sorbent cost/yr
Proactive cleaning	Schedule cleaning to solar peak (zero capture cost)	+2–4% uptime
Battery-forecast LP	Re-solve at day-start with actual SOC	+4–7% battery utilisation
Greedy fallback	No scipy → rule-based schedule	−0 (graceful degradation)

6.2 Seasonal Performance

Season	Night temp range	Capture factor	Net-neg ratio	Notes
Winter (Dec–Feb)	2–15°C	105%	91–93%	Best performance — cool nights, max capture
Spring (Mar–May)	10–25°C	98%	89–91%	Shoulder season — good all-round
Summer (Jun–Sep)	28–38°C	72%	82–87%	Worst — hot nights, capacity penalty
Monsoon (Jul–Sep)	28–35°C	78%	85–88%	Humidity bonus partially offsets heat penalty
Fall (Oct–Nov)	12–28°C	96%	89–91%	Second-best — recovering from summer

Phoenix July nights average 35°C / 95°F — the single largest performance risk. The system correctly penalises capture at high night temperatures and shifts scheduling weight toward shoulder seasons. The 85% net-negative gate is met on an annual average even with summer degradation.

6.3 Sorbent Life Projection

At 0.045% capacity loss per cycle and 3 cycles/day, the sorbent reaches the 60% replacement floor in approximately **2,200 cycles (~24 months)**. Dynamic temperature management (reducing regen temp from 95°C to 82°C as the sorbent ages) extends this to ~27 months without hardware modification. The economic replacement trigger fires earlier when lost revenue from degraded capacity exceeds the amortised cost of a new bed at the prevailing carbon credit price.

7. Cash Flow Projections & Funding Plan

All projections use conservative mid-point assumptions. The 'commercial' column excludes grants — that is the stress-test view of whether the business stands on its own unit economics. **Model the worst plausible month, not the best.**

7.1 Phase-by-Phase Financials

Phase	Duration	Gross CO■ t/mo	Net CO■ t/mo	Revenue $/mo (avg)	Costs $/mo (avg)	Net CF $/mo (avg)	Breakeven
Phase 0	6 mo	0.0	0.0	$0	$850	$-850	—
Phase 1	12 mo	0.1	0.1	$0	$5,257	$-5,257	—
Phase 2	18 mo	25.0	22.0	$7,518	$18,779	$-11,261	grant-dep.
Phase 3	24 mo	250.0	225.0	$106,898	$60,133	$46,764	self-fund
Phase 4	36 mo	100.0	92.0	$49,864	$22,964	$26,900	self-fund

7.2 Revenue Streams

Revenue stream	When	Rate (base case)	Notes
Aggregate sales	Phase 2+	$38–50/t aggregate	ASTM C33 required. ~4.2 t aggregate per t CO■.
Voluntary carbon credits	Phase 2+	$250/t net CO■	Conservative; market range $200–500.
§45Q tax credit	Phase 2+	$130/t (utilisation)	Mineral carbonation = utilisation rate, not storage rate. Verify with couns
DOE SBIR Phase I	Phase 1	~$150k / 30 mo	Non-dilutive. Eligibility: small business, novel tech.
DOE SBIR Phase II	Phase 2	~$1.5M / 24 mo	Conditional on Phase I results.
DOE Carbon Negative Shot	Phase 3+	Prize TBD	$3.5B program, $100/tonne target by 2032.
AZ State / county	Phase 2+	$50–200k/yr	AZ Commerce Authority clean energy grants.

7.3 Summary Financials

$4,980,957
Total revenue (96 mo)
incl. grants

$4,495,965
Commercial revenue
excl. grants

$484,992
Total grants
non-dilutive

$2,676,105
Total costs
all phases

$8,187
Peak cash need
max deficit

Mo 42
Breakeven (commercial)
no grants

7.4 Funding Plan

Phase	Funding source	Amount	Type	Trigger
0	Founder (sweat equity)	$0–2k	Self	Start immediately
1	DOE SBIR Phase I	~$150k	Grant	Company formed, FTO cleared
1	University partnership	$0 cash	In-kind lab access	MOA with university partner
2	DOE SBIR Phase II	~$1.5M	Grant	Phase I complete, bench results pass gate
2	Arizona Commerce Authority	~$100k	Grant	Company in Arizona, pilot operational
3	DOE Carbon Negative Shot	TBD	Prize/contract	Phase 2 gate passed, net-neg verified
3	Aggregate offtake advance	~$50–150k	Revenue advance	ASTM C33 passed, buyer committed
4	Series A (optional)	$1–5M	Equity	Only if replication requires capital

Key constraint: the founder is capital-light and proceeding alongside full-time work and school. Phase 0 has near-zero cash cost. Phase 1 SBIR bridges to Phase 2 without equity dilution. Equity is a last resort — only if scale-out speed requires it.

8. Risk Register & Mitigation

Risk	Likelihood	Impact	Mitigation
Sorbent EF is 15 kg CO■e/kg (high end) → net-neg ratio drops to ~75%	Medium	High	Negotiate supplier EPD in Phase 1. Monte Carlo shows P(below 85%) = 4
Phoenix summer nights >38°C more frequent (climate trend)	Medium	Medium	System already models 35°C baseline. LP scheduler shifts capture weight
Tailings TCLP failure (blocking commercial sale) at primary source	Low-Medium	High	≥2 qualified sources required before Phase 2 gate. Sierrita pilot already fai
§45Q eligibility disputed (mineral carbonation classification)	Low	High	Engage tax counsel before Phase 2. EOR exclusion confirmed in IRA 202
Sorbent IP is not novel / FTO issue	Medium	High	Phase 0 FTO check is a hard gate before any IP investment. Do not file pa
Dust fouling worse than modelled (haboob frequency increasing)	Low-Medium	Medium	Proactive cleaning scheduling already in LP. Filter replacement cost is trac
Carbon credit market price falls below $100/t	Low	Medium	§45Q provides a $130/t floor independent of voluntary market. Aggregate r
Single-founder risk (execution dependency)	High	High	University partnership provides continuity. Document all processes in this c

9. IBM Partnership Proposal

The prototype is ready for Phase 1 IBM deployment today. All code is Python 3.12 with four runtime dependencies. The 11 modules are independently containerisable. A live demo runs at http://localhost:8501 and can be deployed to IBM Cloud in under one business day.

9.1 IBM Platform Architecture

Module	IBM Service	Purpose
Hourly scheduler (optimizer.py)	IBM Cloud Kubernetes Service (IKS)	CronJob pod, daily 06:00
Weather + forecast (weather.py)	watsonx.ai Granite-TTM	72-h PV & temperature forecast
Sensor ingestion (sensors.py)	IBM App Connect	MQTT/OPC-UA → MRV registry
Ledger + MRV (ledger.py, mrv.py)	IBM Cloud Databases for PostgreSQL	Persistent signed records
LCA + Monte Carlo	Watson Studio / Cloud Pak for Data	AutoAI continuous improvement
Dashboard (app.py)	IKS + IBM Cloud Internet Services	Continuous, public URL
Monitoring	IBM Instana	4 custom metrics: net-neg ratio, SOC, dust, sorbent cap
Secrets & config	IBM Secrets Manager	API keys, sensor credentials
Granite-3 assay reader	watsonx.ai LoRA fine-tune	XRF PDF → TailingsAssay JSON (Phase 4)

9.2 Phased IBM Engagement

Phase	Timeline	IBM Services	Deliverable
1 — Infrastructure	Weeks 1–4	IKS, Instana, Cloud Databases, Secrets Manager	All 11 modules containerised, deployed, monitored. 4 custom Insta
2 — Live Data	Weeks 5–8	watsonx.ai Granite-TTM, App Connect	Real NOAA weather → Granite-TTM 72h forecast replacing synthe
3 — AutoAI	Weeks 9–16	Cloud Pak for Data, Watson ML, Watson Studio	30 days of operational data → AutoAI pipeline. Deployed WML mo
4 — Granite Assay	Weeks 17–24	watsonx.ai LoRA, Cloud Object Storage	Fine-tuned Granite-3 8B reads XRF lab PDFs → structured Tailing

9.3 IBM Alignment — Why This Project

- **Net-zero commitment:** IBM targets net-zero by 2030. This project provides verified, permanent, third-party-auditable CO■ removal — not offsets — with a complete and public ledger. IBM could include verified credits from this system in its own portfolio.
- **Referenceable watsonx sustainability demo:** A system where IBM technology directly enables a verifiable, permanent removal credit — with Granite-TTM forecasting, App Connect sensor integration, and AutoAI scheduling improvement — is a concrete demonstration of watsonx's sustainability use case.
- **Arizona copper mine opportunity:** Arizona mines generate >200 Mt tailings/yr. At full DAC-scale utilisation, this is a 20 Mt CO■/yr removal opportunity — within the range of national climate targets. IBM's Arizona cloud infrastructure (Phoenix data center) is geographically aligned.
- **IP-light engagement:** The prototype is open to co-development. IBM's contribution is infrastructure and AI services; the carbon accounting methodology and MRV registry are the founder's IP. No IP conflict.

9.4 Requested from IBM Client Engineering

- 4-week IBM Client Engineering engagement to complete Phase 1 (IKS deployment, Instana monitoring, PostgreSQL integration) and validate against IBM Cloud production standards.
- Access to watsonx.ai Granite-TTM time series models for weather forecast integration under a watsonx trial or co-development agreement.
- Introduction to IBM's sustainability practice and carbon market advisory resources to structure the credit issuance and third-party verification pathway.
- Joint go-to-market exploration for watsonx.ai + DAC as a referenceable IBM sustainability demonstration project with potential inclusion in IBM's own carbon removal portfolio.

10. Appendices

Appendix A — Emission Factors

Factor	Value	Unit	Source	Confid
Solar PV lifecycle	0.020	kg CO■e/kWh	IPCC AR6 WG3 Table A.III.2	High
Battery mfg overhead	0.008	kg CO■e/kWh discharged	Ellingsen et al. 2022, LFP pack	Medium
Sorbent manufacturing	4–15 (mode 8)	kg CO■e/kg	Fasihi 2019; Wurzbacher 2016	LOW ■
Diesel crusher	0.27	kg CO■e/kWh	US EIA 2023 diesel genset	High
Solar crusher	0.020	kg CO■e/kWh	Same as PV lifecycle	High
Truck transport	0.062	kg CO■e/t·km	Arizona DOT 2022; IPCC AR6 T10.SM.11	High
Infrastructure (annualised)	1–4 (mode 2)	kg CO■e/t gross CO■	Climeworks 2021 LCA supp.	LOW ■
Filter manufacturing	8–28 (mode 15)	kg CO■e/set	Ecoinvent 3.9 HEPA	LOW ■
Municipal water	0.00035	kg CO■e/L	WRI Aqueduct; AZ municipal treatment	Medium
Geologic injection	0.040	kWh/kg CO■	IEA injection energy estimate	Medium

Appendix B — Arizona Mine Database

Mine	Operator	Mineral type	Reactivity	Distance	Grinder	Notes
Morenci	Freeport-McMoRan	Olivine/pyroxene	18–22%	230 km	Diesel	World's largest open-pit copper mine. High c
Bagdad	Freeport-McMoRan	Serpentinite	22–28%	310 km	Diesel/Solar	Highest reactivity. Softest ore — lowest grin
Sierrita	Freeport-McMoRan	Wollastonite	8–14%	90 km	Diesel	Closest mine, lowest reactivity. Pilot lot faile
Ray	Asarco	Mixed silicate	12–18%	110 km	Diesel	Moderate reactivity and proximity. Good bal
Pinto Valley	BHP	Chlorite/talc	10–16%	150 km	Solar	Solar crusher on-site. Lowest grind energy.

Appendix C — Glossary

Term	Definition
DAC	Direct Air Capture — mechanical removal of CO■ directly from ambient air
Net-negative	System removes more CO■ than it emits across its full lifecycle
Gross tonne	CO■ adsorbed by the sorbent — the vendor metric (incomplete)
Net tonne	CO■ removed after all system emissions are deducted — the honest metric
MRV	Measurement, Reporting, Verification — the auditable chain from sensor to credit
TCLP	Toxicity Characteristic Leaching Procedure — EPA test for hazardous waste leaching
ASTM C33	Standard Specification for Concrete Aggregates (ASTM International)
§45Q	IRS Section 45Q tax credit for carbon capture and storage ($130/t utilisation, $180/t storage)

Term	Definition
LCA	Life Cycle Assessment — accounting for all emissions across a product's lifecycle
Bond's Law	Grinding energy equation: W = Wi × 10 × (1/√P■■ – 1/√F■■) in kWh/t
EOR	Enhanced Oil Recovery — CO■ injected to extract more oil. NOT a permanent fate.
FTO	Freedom to Operate — patent search to confirm no blocking IP
SBIR	Small Business Innovation Research — US federal grant program (DOE, NSF, etc.)
GHI	Global Horizontal Irradiance — solar resource measurement (W/m²)
LFP	Lithium Iron Phosphate — battery chemistry (high cycle life, thermal stability)

Appendix D — Software Files

File	Lines (approx.)	Purpose
weather.py	~250	Synthetic Phoenix weather generator
sorbent.py	~170	Amine sorbent degradation + replacement economics
tailings.py	~420	Per-batch assay engine, Bond's Law, mine database
ledger.py	~250	Net-negative carbon ledger, all emission debits
optimizer.py	~450	24h LP scheduler + 5 optimisations
mrv.py	~430	MRV batch traceability chain, TCLP gates
lca.py	~310	ISO 14064-2 LCA boundary, water accounting
phase_gate.py	~310	Phase 0–4 go/no-go engine
cashflow.py	~420	96-month revenue/cost model, §45Q, grants
sensors.py	~390	Sensor pipeline, QC, auto-MRV commit
monte_carlo.py	~265	Monte Carlo LCA sensitivity, tornado chart
app.py	~1,400	11-tab Streamlit dashboard (main UI)
demo.py	~200	CLI demo (no web UI required)
build_pdf.py	~820	This document generator
requirements.txt	5	streamlit, plotly, pandas, scipy, reportlab